

Haryana LEET / OCET 2026

Section A — Basic Sciences

25 Questions | 25 Marks | No Negative Marking

a-1 MATHEMATICS (8 Q | 8 M)

■ Arithmetic Progressions (AP)

- nth term: $a + (n-1)d$ | Sum: $n/2[2a+(n-1)d]$
- AP in word problems — very frequently asked
- Insert arithmetic means between two numbers

■ Matrices

- Types: row, column, square, identity, zero matrix
- Addition, subtraction, multiplication of matrices
- Transpose, symmetric, skew-symmetric matrices
- Determinant of 2×2 & 3×3 matrix | Inverse of 2×2

■ Complex Numbers

- $i = \sqrt{-1}$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
- Modulus $|z|$, Argument, Conjugate, Polar form
- Addition, multiplication, division of complex numbers

■ Logarithm

- $\log(mn) = \log m + \log n$ | $\log(m/n) = \log m - \log n$
- $\log(m)^n = n \log m$ | Change of base formula

■ Permutation & Combination

- $nPr = n!/(n-r)!$ | $nCr = n!/[(r!(n-r)!)]$
- Word arrangements, selection problems

■ Binomial Expansion

- General term: $T(r+1) = nCr \cdot a^{n-r} \cdot b^r$
- Middle term, independent term, coefficient of x^n

■ Probability

- $P(A) = \text{favourable}/\text{total}$ | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Conditional probability $P(A|B)$ | Independent events
- Complementary events: $P(A') = 1 - P(A)$

■ Trigonometry

- Standard values: $\sin/\cos/\tan$ at $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$
- Identities: $\sin^2\theta + \cos^2\theta = 1$, $\sec^2\theta = 1 + \tan^2\theta$, $\text{cosec}^2\theta = 1 + \cot^2\theta$
- Compound angle: $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$

■ Straight Line & Circle

- Slope: $m = (y_2 - y_1)/(x_2 - x_1)$ | Slope-intercept: $y = mx + c$
- Distance between two points | Midpoint formula
- Circle: $(x-h)^2 + (y-k)^2 = r^2$ | General equation

■ Differentiation

- $d/dx(x^n) = nx^{n-1}$ | Product, Quotient, Chain rule
- Successive differentiation | Second derivative
- Maxima & Minima: $dy/dx = 0$ and sign of d^2y/dx^2

■ Integration

- $\int x^n dx = x^{(n+1)}/(n+1) + C$ | Standard integrals
- Definite integral & area under curve
- First order first degree ODE — variable separable method

■ Statistics

- Mean (AM), Median, Mode — formulas & problems
- Mean Deviation from mean/median
- Standard Deviation: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2]/n}$
- Variance = σ^2

a-2 PHYSICS (8 Q | 8 M)

■ Units & Dimensions

- SI base units — 7 fundamental quantities & units
- Dimensional analysis: checking equations, deriving formulas
- Limitations: no info about physical constants, dimensionless quantities

■ Motion in 1D & Newton's Laws

- $v = u + at$ | $s = ut + \frac{1}{2}at^2$ | $v^2 = u^2 + 2as$
- Newton's 3 laws — statement & applications
- Friction: static, kinetic, rolling | $f = \mu N$

■ Work, Energy & Conservation Laws

- $W = F \cdot d \cdot \cos\theta$ | $KE = \frac{1}{2}mv^2$ | $PE = mgh$
- Work-energy theorem: $W_{\text{net}} = \Delta KE$
- Conservation of mechanical energy — no friction problems

■ Rotational Motion

- Torque $\tau = r \times F$ | Moment of inertia $I = \sum mr^2$
- $L = I\omega$ | Conservation of angular momentum
- Rolling motion: $KE = \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$

■ Waves & SHM

- SHM: $x = A \sin\omega t$ | $T = 2\pi\sqrt{m/k}$ for spring
- Characteristics: amplitude, frequency, time period, phase
- Transverse & longitudinal waves | Wave speed $v = f\lambda$

■ Properties of Matter

- Elasticity: stress, strain, Young's modulus $Y = \text{stress/strain}$
- Surface tension: $S = F/L$ | Capillary rise $h = \frac{2S \cos\theta}{r \rho g}$
- Viscosity: $\eta = F \cdot d / (A \cdot v)$ | Stoke's law

■ Heat & Temperature

- Temperature scales: °C, K, °F conversions
- Modes of heat transfer: Conduction (Fourier's law), Convection, Radiation
- Newton's law of cooling | Stefan-Boltzmann law

■ Geometric Optics

- Mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ | $m = -v/u$
- Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ | Lens maker's equation
- Refraction: Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ | TIR

■ Electrostatics

- Coulomb's law: $F = kq_1q_2/r^2$ | $k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$
- $E = kq/r^2$ | $V = kq/r$ | Relation $E = -dV/dr$
- Capacitor: $C = Q/V$ | Parallel plates: $C = \epsilon_0 A/d$

- Dielectric constant K: $C_{\text{new}} = KC$

■ Modern Physics

- Laser: stimulated emission, population inversion, coherence
- Uses of laser: surgery, cutting, CD/DVD, range-finding
- Superconductivity: zero resistance below critical temp
- Non-conventional energy: solar, wind, tidal, geothermal

a-3 CHEMISTRY (3 Q | 3 M)

■ Water & Hardness

- Hard water: contains Ca^{2+} , Mg^{2+} salts — won't lather with soap
- Temporary hardness: $\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$ — removed by boiling
- Permanent hardness: sulphates/chlorides — removed by chemical methods
- Ion exchange method: zeolite / synthetic resin process

■ pH, Acidity & Basicity

- $\text{pH} = -\log[\text{H}^+]$ | $\text{pH} < 7$ acidic | $\text{pH} = 7$ neutral | $\text{pH} > 7$ basic
- Acidity: no. of replaceable H^+ ions | Basicity: no. of replaceable OH^- ions

■ Electronic Configuration

- Aufbau principle, Hund's rule, Pauli exclusion principle
- Must know config of elements 1-30 (H to Zn)
- Example: $\text{Fe}(26) = [\text{Ar}] 3d^6 4s^2$ | $\text{Cu}(29) = [\text{Ar}] 3d^5 4s^1$

■ Basic Definitions

- Equivalent weight = Mol. wt / valency (for acids/bases)
- Chemical equation: law of conservation of mass
- Symbol, Formula, Valency — IUPAC basics

a-4 COMMUNICATION SKILLS (3 Q | 3 M)

■ Vocabulary

- Synonyms & Antonyms — 500+ common words list
- One-word substitution

■ Grammar — High Priority Topics

- Tenses: 12 tenses — identification & conversion
- Active $\leftrightarrow$ Passive Voice — all tenses
- Direct $\leftrightarrow$ Indirect Speech (Narration changes: tense, pronoun, time)
- Prepositions: in/on/at/by/with/for/since/from — correct usage

■ Sentence Correction

- Subject-verb agreement errors
- Punctuation: comma, full stop, apostrophe, colon
- Correction of incorrect sentences

a-5 GENERAL AWARENESS (3 Q | 3 M)

■ Haryana Technical Education Facts

- Total polytechnics in Haryana & their sanctioned intake
- LEET/OCET eligibility criteria (45% diploma / 40% reserved)
- Number of B.Tech & B.Pharmacy institutes in Haryana
- Steps of online off-campus counselling process

■ Haryana — General Knowledge

- Formation: 1 November 1966 (carved from Punjab)
- Districts: 22 | Tehsils: 93 | Sub-Tehsils: 50+
- Area: 44,212 km² | Population: ~2.8 crore (2011 census)
- Boundary states: Punjab (N/W), Himachal Pradesh (N), Uttarakhand (NE), UP (E), Rajasthan (S/W), Delhi (E)
- Capital: Chandigarh (shared with Punjab)
- State animal: Blackbuck | State bird: Black Francolin (Kala Teetar)